

Dataprep Regex Cheat Sheet – Replace Invalid Data (Email, Phone, SSN & More)



FEC 2016 >
Campaign Contributions 2016 – 2 > Initial sample

column2

column9

column8

#

sum_column16

#

average_column16

#

19 Categories

7,693 Categories

2 Categories

-5.4k - 10.8k

-5,400 - 10,800

0 - 6

C00573519

BALDWIN, JONATHAN

IND

25

25

C00573519

WOOD ZENTHOEFER, KATHRYN DR.

IND

1350

1350

C00573519

ADAMSON, ANN MRS.

IND

100

100

C00573519

FORD, JIMMY

IND

25

25

C00573519

HOLIAN, CHRIS

IND

19

19

C00573519

SASSER, DONNA

IND

25

25

C00573519

WEAVER, SANDRA C. MRS.

IND

100

100

C00573519

BROOKE, SHIRLEY G. MS.

IND

100

100

C00573519

ENGLAND, RUTH M. MRS.

IND

100

100

C00573519

RABITO, MARK J. MR.

IND

500

500

C00574624

BYLER, CHRISTINE RAE

IND

50

50

C00574624

ALESHIRE, SUSAN

IND

200

200

C00574624

CAMPBELL, PHYLLIS

IND

100

100

C00574624

SKARBOSZEWSKI, VICKI

IND

150

75

C00574624

DINSMORE, DAVID MR.

IND

50

50

C00574624

KUROWSKI, KENNETH J. MR.

IND

25

25

C00574624

PHILLIPS, KAREN

IND

200

200

C00574624

SPARKS, ROBERT

IND

15

15

C00575795

NEYON, DAVID

IND

250

250

C00577130

SCHIDE, ROBERT

IND

100

100

C00577130

BAZNIK, CONNIE

IND

50

50

C00577130

HAMILTON, AARON

IND

250

250

C00577130

MCILLAN, KENT

IND

60

60

C00573519

BAUMAN, PHILIP Q. MR.

IND

25

25

C00573519

BEDNARZ, JEAN

IND

25

25

New Step

Recipe

1 Replace matches of '(start)'(end)' from all columns with ''

2 Inner join with Candidate Master 2016 – 2 on column2 == column11

3 Pivot and compute 3 functions grouped by 3 columns

6 Columns

7,712 Rows

3 Data Types

The screenshot shows the Google Cloud Dataprep (Trifacta) interface. The main table has columns 'order_date' and 'product_category'. A context menu is open over the 'order_date' column, with the 'Text or pattern' option highlighted. The right sidebar shows details for the 'order_date' column, including a quality bar (100% valid), unique values, distribution, and patterns.

4

This is a **complete, practical, interview-ready cheat sheet** for **Google Cloud Dataprep (Trifacta)** to **identify and replace invalid values using regex**.

You can use these in:

- **Replace values** → **Custom text or pattern**
- **Conditional replace**
- **Formula builder (REGEXMATCH)**



GENERAL PATTERN (IMPORTANT)

✓ **Valid pattern**

`^VALID_REGEX$`

✗ Invalid pattern (used for replacement)

```
^(?!VALID_REGEX).*
```

➔ This **negative lookahead** matches **only invalid values**.

EMAIL

✓ Valid Email Regex (Practical)

```
^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$
```

✗ Replace Invalid Emails

```
^(?![A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$).*
```

Replace with:

NULL or INVALID_EMAIL

PHONE NUMBER

Indian Mobile (10 digits, optional +91)

✓ Valid

```
^(\\+91[- ])?[6-9]\\d{9}$
```

✗ Invalid

```
^(?!(\\+91[- ])?[6-9]\\d{9}$).*
```

International Phone (E.164)

✓ Valid

`^\+?[1-9]\d{7,14}$`

✗ Invalid

`^(?!^\+?[1-9]\d{7,14}$).*`



SSN (US)

✓ Valid SSN

`^\d{3}-\d{2}-\d{4}$`

✗ Invalid SSN

`^(?!^\d{3}-\d{2}-\d{4}$).*`



AADHAAR (India)

✓ Valid Aadhaar (12 digits)

`^\d{12}$`

✗ Invalid Aadhaar

`^(?!^\d{12}$).*`



PAN (India)

✓ Valid PAN

`^[A-Z]{5}[0-9]{4}[A-Z]$`

✗ Invalid PAN

`^(?![A-Z]{5}[0-9]{4}[A-Z]$).*`

URL

✓ Valid URL

`^(https?:\/\/)?([\w\-\.]+\.[a-zA-Z]{2,}\.\/?).*`

✗ Invalid URL

`^(?!((https?:\/\/)?([\w\-\.]+\.[a-zA-Z]{2,}\.\/?).*$)).*`

PIN / ZIP CODE

India PIN

✓ Valid

`^\d{6}$`

✗ Invalid

`^(?!\d{6}$).*`

US ZIP

✓ Valid

`^\d{5}(-\d{4})?$`

 Invalid

`^(?!\\d{5})(-\\d{4})?$).*`



DATE (yyyy-mm-dd)

 Valid Date

`^\\d{4}-\\d{2}-\\d{2}$`

 Invalid Date

`^(?!\\d{4}-\\d{2}-\\d{2}$).*`



NUMERIC ONLY

 Valid

`^\\d+$`

 Invalid

`^(?!\\d+$).*`



ALPHABETIC ONLY

 Valid

`^[A-Za-z ]+$`

 Invalid

`^(?![A-Za-z ]+$).*`



DATAPREP FORMULA (SAFE METHOD)

Replace invalid values with NULL

```
IF(  
  REGEXMATCH(column_name, 'VALID_REGEX'),  
  column_name,  
  NULL  
)
```

Example (Email)

```
IF(  
  REGEXMATCH(email,  
  '^ [A-Za-z0-9._%+- ]+@[A-Za-z0-9.- ]+\.[A-Za-z]{2,}$'),  
  email,  
  NULL  
)
```



ENTERPRISE BEST PRACTICE

- ✓ Replace invalid values → NULL
- ✓ Add **DQ flag** column

```
IF(REGEXMATCH(email, 'VALID_REGEX'), 'VALID', 'INVALID')
```

✓ Track:

- % invalid emails
- % invalid phones
- % invalid IDs

INTERVIEW-READY ONE-LINER

In Dataprep, invalid data is replaced using regex-based conditional transforms, typically leveraging negative lookahead patterns to identify non-conforming values and standardize them to NULL or quality flags.

- Convert these rules to **BigQuery SQL**
- Provide **Dataprep UI screenshots**
- Build a **Data Quality rulebook (PDF)**
- Give **certification MCQs**
- Show **Dataflow regex validation**